

Autonomous SysAdmin – Agentic AI + Systems

Team:

Epari Subhransi

Dhriti Kamani

Deeptha S

Mufaddal Agarwala

Mallikarjuna Rao R.V

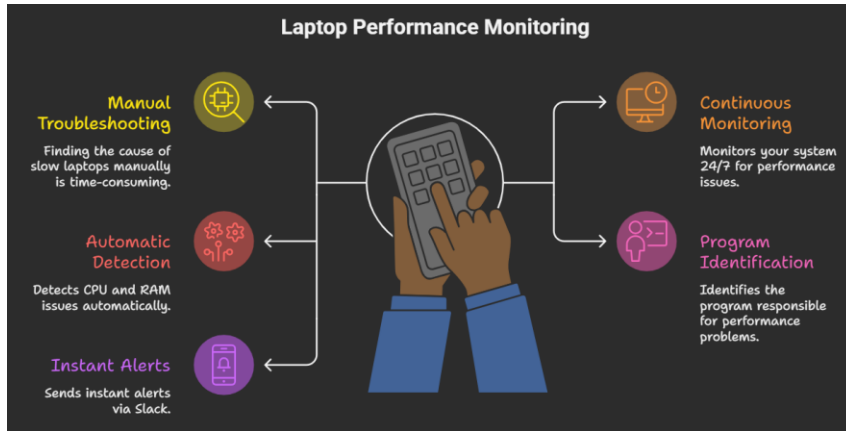
Nishitha Sathish

Centre for Innovation and Entrepreneurship (CIE)



Laptops slow down unexpectedly, and finding the cause manually is time-consuming

- Monitors your system 24/7
- Detects CPU/RAM issues automatically
- Identifies the responsible program
- Sends instant alerts via Slack



NOW: THE FRUSTRATION.

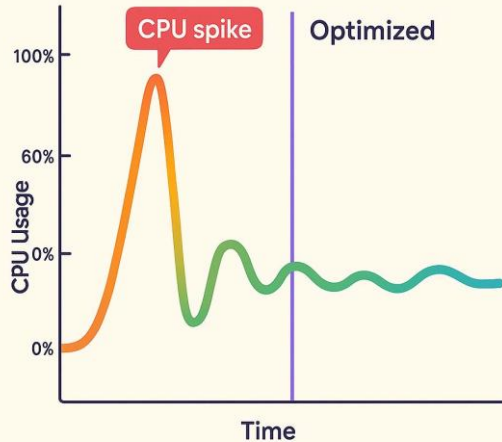
Right now, you'd have to manually check what's eating up your CPU or RAM — opening Task Manager, scanning processes, googling the problem.

That's exactly what this project builds.



The Problem We're Solving

Server crashes cost businesses Rs 5,600/minute on average. Manual detection + diagnosis = too slow, too human-dependent.



Detection Lag

- Late issue detection
- System already down
- ⌚ 5–30 min delay

Diagnosis Complexity

- Needs Linux expertise
- Hard to find culprit
- ⌚ Hours to debug

Cost of Inaction

- No SRE = higher risk
- Frequent outages
- 💰 Rs100K+/hr loss

SysAdmin

Sense the glitch. Name the culprit. Solve the mystery.

Click anywhere to open

Three wavy, light blue lines that span the width of the page, positioned below the text 'Click anywhere to open'.



Data Layer (psutil)

- Samples CPU every 5s
- Triggers at CPU > 80% / RAM > 90%
- Sends structured events
- Extendable (disk, network, logs)



Intelligence Layer (LLM)

- Writes final summary only
Logic = deterministic
- Input: structured system data
- Output: 3-line RCA



Python + psutil System Monitoring Layer

- ▶ Cross-platform CPU/RAM/Disk stats
- ▶ Process-level details (PID, name, %CPU)
- ▶ Lightweight, zero overhead

CrewAI Agent Orchestration

- ▶ Role-based agent personas (Watcher, Detective, Reporter)
- ▶ Structured reasoning chains
- ▶ LangSmith for tracing agent decisions

OpenAI / Llama 3 LLM (Report Generation)

- ▶ GPT-4o or Llama 3 via Ollama (local)
- ▶ Structured prompt for 3-sentence RCA
- ▶ Graceful API failure handling

Slack Webhooks Notification Layer

- ▶ Instant formatted alerts with PID, RCA, recommendations
- ▶ Swappable for Discord, Email, PagerDuty
- ▶ JSON payload, isolated module

Docker Containerization

- ▶ Isolated environment, no host system risk
- ▶ Portable — runs on any machine
- ▶ Demo-ready with one docker run command

YAML Config + JSON Logs Config & Observability

- ▶ settings.yaml: thresholds, webhook URLs, env-specific config
- ▶ Structured JSON logs: ELK-ready for future dashboards
- ▶ Change thresholds without touching code

Watching quietly. The app checks CPU, memory, disk, and running apps in the background.

AI: Ready (CrewAI)

Watching

Checking

Writing report

Done

CPU Usage

35.9%

CPU Usage

16 cores

Memory (RAM)

90.5%

Memory (RAM)

14.2 / 15.7 GB

Disk Space C:

98.6%

Disk Space C:

269.3 / 273.1 GiB

How full your main drive (C:) is.
Above 95% = nearly full.

Process Load

100.0%

Process Load

371 processes (threshold 300)

AI Agent Live Thinking

Watching quietly

Time Detail

09:28:23 [INFO] Prometheus exporter
listening on :910809:28:23 [OK] Thresholds: CPU>80% RAM>85%
Disk>95%09:28:23 [WARN] Disk C: is at 98.5% consider freeing space
soon.09:28:35 [ERR] INCIDENT -
MEMORY_SPIKE

Active Apps (by CPU usage)

The list updates continuously and filters Windows system pseudo-processes.

PID	Name	CPU %	Mem %
40688	ollama.exe	389.4	3.3
23372	python.exe	84.0	1.6
23592	wpp.exe	11.3	2.1
38704	dwm.exe	9.2	0.8

Investigation finished. The AI is writing a plain-English report.

AI: Writing report

Watching

Checking

Writing report

Done

CPU Usage

19.8%

CPU Usage

16 cores

Memory (RAM)

97.1%

Memory (RAM)

15.2 / 15.7 GB

Disk Space C:

98.6%

Disk Space C:

269.3 / 273.1 GiB

Process Load

100.0%

Process Load

374 processes (threshold 300)

AI Agent Live Thinking

AI is writing the report...

Time Detail

- 09:28:35 ⚙️ Trigger [MEMORY_SPIKE] – handing off to local AI agent...
- 09:28:35 🔍 Detective agent starting diagnostic chain...
- 09:30:44 💡 Local AI is analyzing your system – this may take 15–30 seconds...
- 09:32:45 ⚠️ Analysis is taking longer than expected. Preparing a quick local report.

Time

Level

- 09:28:23 [INFO] Autonomous SysAdmin started watching your system.
- 09:28:23 [INFO] Prometheus exporter listening on :9108
- 09:28:23 [OK] Thresholds: CPU>80% RAM>85% Disk>95%
- 09:28:23 [WARN] Disk C: is at 98.5% consider freeing space soon

Active Apps (by CPU usage)

The list updates continuously and filters Windows system pseudo-processes.

PID	Name	CPU %	Mem %
23372	python.exe	101.5	1.4
38704	dwm.exe	20.1	1.2
41480	msedge.exe	16.6	2.0
30024	MsMpEng.exe	7.9	2.0

Problem Found

Here's What the AI Discovered

Alert PID: 18768

Memory exhaustion (92.6%) system approaching OOM. Diagnostic tools identified PID N/A as the root cause abnormal resource consumption and excessive handle acquisition detected. Recommended action: terminate PID N/A and monitor system recovery.

Kill Process

Reset

Slack: send failed

AUTONOMOUS SYSADMIN AI-powered health monitor for your Windows machine

LIVE

Running low on memory

running for 00:04:43 time 09:33:24

The AI is actively diagnosing the issue right now.

AI: Investigating

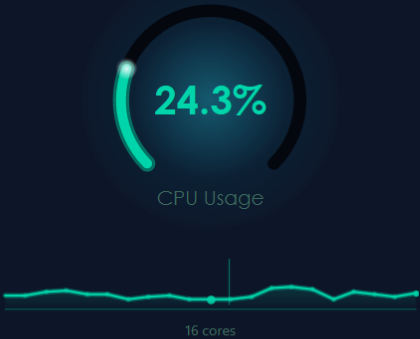
Watching

Checking

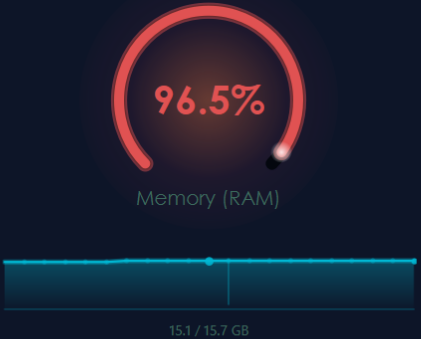
Writing report

Done

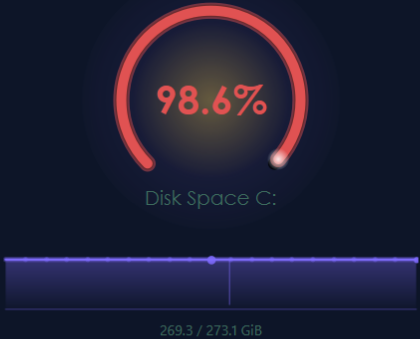
CPU Usage



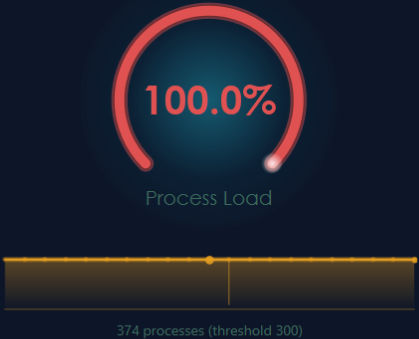
Memory (RAM)



Disk Space C:



Process Load



AI Agent Live Thinking

AI is investigating

your system

Active Apps (by CPU usage)



5-30 min

saved per incident



Rs 100K+/hr

downtime cost avoided



0 humans

needed to diagnose



<60 sec

from spike to Slack alert



For Small Companies

No dedicated SRE team? No problem.

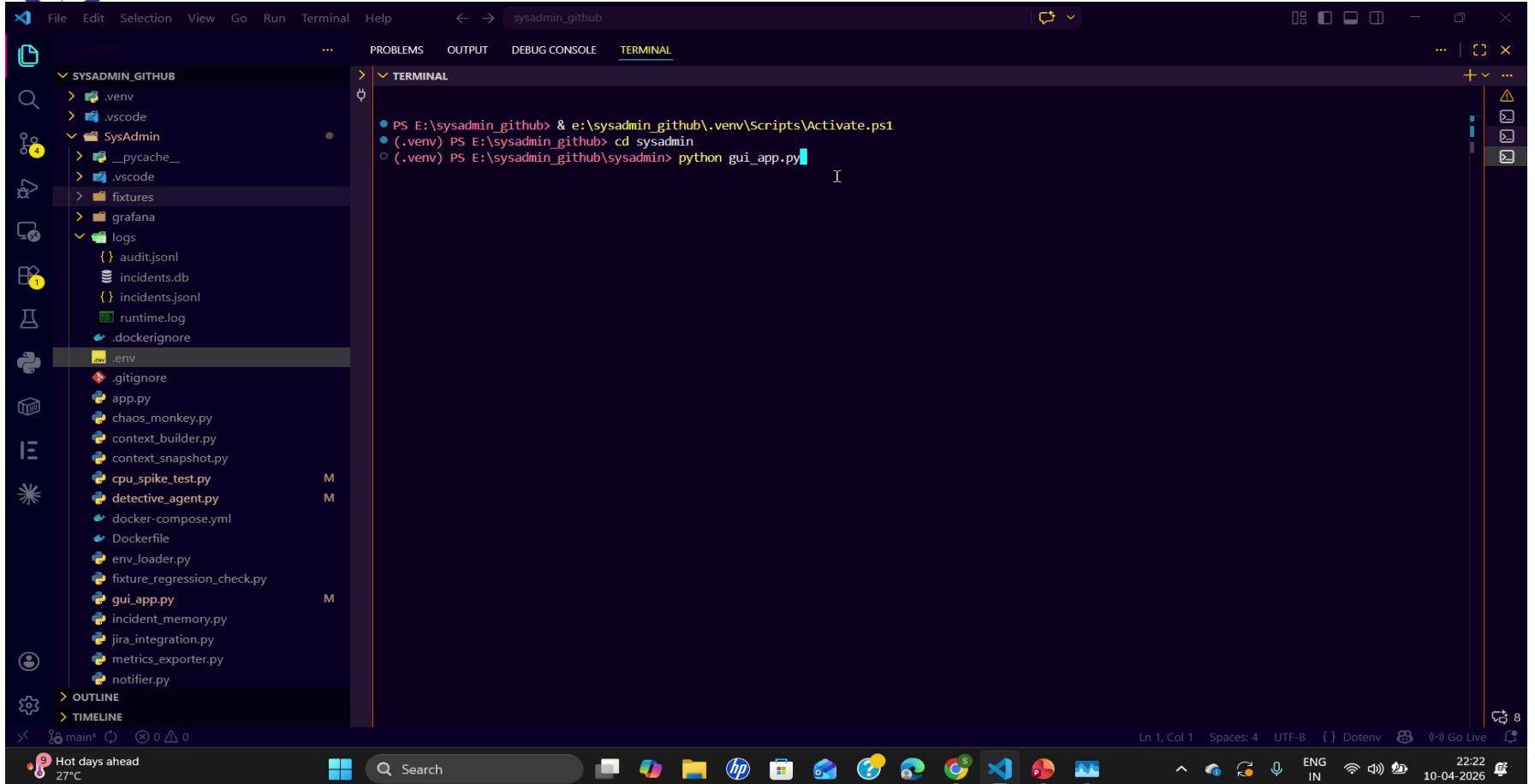
- High outage risk
- Slow recovery



As a Learning Project

Shows senior-level thinking at student level.

- Real-world system design
- AI + DevOps skills
- Strong resume project



The screenshot shows a Visual Studio Code editor window with a terminal open. The left sidebar displays the file explorer for a project named 'sysadmin_github'. The terminal window shows the following commands and output:

```
PS E:\sysadmin_github> & e:\sysadmin_github\.venv\Scripts\Activate.ps1
(.venv) PS E:\sysadmin_github> cd sysadmin
(.venv) PS E:\sysadmin_github\sysadmin> python gui_app.py
```

The file explorer on the left shows the following structure:

- sysadmin_github
 - .env
 - .vscode
 - SysAdmin
 - __pycache__
 - .vscode
 - fixtures
 - grafana
 - logs
 - audit.jsonl
 - incidents.db
 - incidents.jsonl
 - runtime.log
 - .dockerignore
 - .env
 - .gitignore
 - app.py
 - chaos_monkey.py
 - context_builder.py
 - context_snapshot.py
 - cpu_spike_test.py
 - detective_agent.py
 - docker-compose.yml
 - Dockerfile
 - env_loader.py
 - fixture_regression_check.py
 - gui_app.py
 - incident_memory.py
 - jira_integration.py
 - metrics_exporter.py
 - notifier.py

The bottom status bar shows the current file is 'main*', the encoding is 'UTF-8', and the language is 'Python'.

Autonomous SysAdmin

Agentic AI + Systems Engineering

👁️ SENSE → 🧠 THINK → ⚙️ ACT → 📢 REPORT

"I detected a RAM spike. Python script (PID 4213) is leaking memory.
I recommend terminating it."

Runs in Docker. No root access.
CIE | PES University